7 Powerful Techniques for Group Theory

Mathematical crowbars that turn locked safes into open doors.

1. Subgroup Test Shortcut

Idea One condition replaces three. A nonempty $H \subseteq G$ is a subgroup iff

$$a, b \in H \implies ab^{-1} \in H.$$

Illustration For a set defined by a divisibility rule, just check that the difference of two elements still satisfies the rule.

Problem Show $H = \{x \in \mathbb{Z} : 4 \mid x\}$ is a subgroup of $(\mathbb{Z}, +)$.
Hint: if $a = 4m$ and $b = 4n$, then $a - b = 4(m - n) \in H$. □

2. Order Divisibility Trick (Lagrange)

Statement For any element a of a finite group G,

$$|a| \mid |G|.$$

Illustration If $|G| = 12$, an element cannot loop back after 5 steps — the cycle length must divide 12.

Problem Let $|G| = 21$. What are the possible orders of elements of G?
Answer: divisors of 21: $1, 3, 7, 21$.

3. Use Cosets to Count

Key Formula The number of distinct left cosets of H in G is

$$[G : H] = \frac{|G|}{|H|}.$$

Illustration A warehouse of 24 crates in stacks of 3 immediately gives 8 stacks — no counting required.

Problem If $|G| = 20$ and $|H| = 5$, how many cosets does H have in G?
Answer: $20/5 = 4$.

4. Normal Subgroup Detection

Useful Conditions $N \trianglelefteq G$ if any one of the following holds:

(i) $gNg^{-1} = N$ for all $g \in G$.

(ii) $gN = Ng$ for all $g \in G$.

(iii) $N = \ker \phi$ for some homomorphism ϕ.

Illustration A normal subgroup is a perfectly centered gear: rotate the whole machine and the gear stays aligned.

Problem Show every subgroup of an abelian group is normal.
Reason: $ab = ba$ implies $gNg^{-1} = N$ for all g. □

5. Generators Strategy

Idea In a cyclic group $G = \langle a \rangle$, every element has the form a^n. Finding the generators reduces the entire group to one element.

Illustration A bicycle pedal: one rotation drives every spoke — one generator produces the whole group.

Problem Find all generators of the cyclic group $\mathbb{Z}_{10}$.
Answer: elements with $\gcd(k, 10) = 1$: $1, 3, 7, 9$.

6. Kernel Hunting

Definition For a homomorphism $\phi : G \to H$,

$$\ker \phi = \{g \in G : \phi(g) = e_H\}.$$

By the **First Isomorphism Theorem**: $G/\ker \phi \cong \phi(G)$.

Illustration The kernel reveals exactly which elements collapse to the identity, exposing the hidden structure of the group.

Problem For $\phi : \mathbb{Z} \to \mathbb{Z}_8$, $\phi(n) = n \bmod 8$, find $\ker \phi$.
Answer: $\ker \phi = 8\mathbb{Z}$.

7. Permutation Decomposition

Idea Every permutation in S_n decomposes uniquely (up to order) into disjoint cycles. The cycle type immediately reveals the order and parity.

Example $(1\ 4\ 2)(3\ 5)$: sends $1 \to 4 \to 2 \to 1$ and $3 \leftrightarrow 5$.
Order $= \operatorname{lcm}(3, 2) = 6$. Two cycles of lengths 3 and 2 $\Rightarrow$ odd permutation.

Problem Find the order and parity of the permutation $(1\ 3\ 5\ 2)(4\ 6)$ in S_6.

Strategic order. (1) Subgroup criterion (2) Lagrange's theorem (3) Coset counting (4) Normality test (5) Generators (6) Kernel (7) Cycle decomposition.
With practice, group theory becomes a map rather than a maze.

Module 1 — Groups

The mathematics of structure and symmetry.

1. Groups as Symmetries

Illustration Rotating a square by 0°, 90°, 180°, 270° always returns it to an indistinguishable position. These four rotations form the symmetry group of the square, the dihedral group D_4.

Definition A **group** is a pair $(G, *)$ where $* : G \times G \to G$ satisfies:

(i) **Closure:** $a, b \in G \Rightarrow a * b \in G$.

(ii) **Associativity:** $(a * b) * c = a * (b * c)$.

(iii) **Identity:** $\exists\, e \in G$ with $a * e = e * a = a$.

(iv) **Inverses:** $\forall\, a \in G$, $\exists\, a^{-1} \in G$ with $a * a^{-1} = e$.

Plain English A group is a system of reversible moves: combine any two moves, stay inside the system, one move does nothing, and every move can be undone.

Problem Show that $G = \{1, -1, i, -i\}$ under multiplication is a group.

2. Important Examples of Groups

Illustration Each classic group governs a different kind of symmetry or transformation:

Group	What it controls
$\mathbb{Z}_n$	clock arithmetic (addition mod n)
S_n	all permutations of $\{1, \ldots, n\}$
A_n	even permutations in S_n
D_n	symmetries of a regular n-gon
$GL(n, \mathbb{R})$	invertible $n \times n$ real matrices
$SL(n, \mathbb{R})$	matrices with determinant 1
$O(n, \mathbb{R})$	orthogonal matrices ($A^T A = I$)
$SO(n, \mathbb{R})$	rotations ($A^T A = I$, $\det A = 1$)

Definition

$$GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det(A) \neq 0\}, \quad SL(n, \mathbb{R}) = \{A : \det(A) = 1\},$$

$$O(n, \mathbb{R}) = \{A : A^T A = I\}, \quad SO(n, \mathbb{R}) = \{A \in O(n, \mathbb{R}) : \det(A) = 1\}.$$

Plain English These groups represent different types of transformations. Modern physics and machine learning rely on several of them.

Problem Find $|S_5|$.

3. Elementary Properties of Groups

Illustration Moves in chess: grouping $(AB)C$ or $A(BC)$ gives the same result. Groups behave like a perfectly designed machine with no ambiguity.

Definition In any group G:

(i) The identity e is unique.

(ii) The inverse a^{-1} of each element is unique.

(iii) $(ab)^{-1} = b^{-1}a^{-1}$.

(iv) **Cancellation:** $ab = ac \Rightarrow b = c$ and $ba = ca \Rightarrow b = c$.

Plain English Groups behave predictably: one neutral element, unique inverses, and algebra works cleanly.

Problem Prove that $(ab)^{-1} = b^{-1}a^{-1}$.
Hint: compute $(ab)(b^{-1}a^{-1})$ and simplify.

4. Abelian Groups

Illustration Integer addition: $3 + 5 = 5 + 3$. The order of operands does not matter. Such groups are called **abelian**, after *Niels Henrik Abel*.

Definition A group $(G, *)$ is **abelian** if

$$a * b = b * a \quad \text{for all } a, b \in G.$$

Plain English Most everyday arithmetic is abelian. Matrix multiplication, however, is not — order matters in $GL(n, \mathbb{R})$.

Problem Is $GL(2, \mathbb{R})$ abelian? Test with

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Compute AB and BA and compare.

Module 1 complete. Next: cosets, Lagrange's theorem, quotient groups, and homomorphisms — where abstract algebra begins to feel like architecture.

Module 2 — Subgroups & Cosets

Rooms, hallways, and blueprints inside a mathematical building.

1. Subgroups and Centralizers

Illustration The even integers $2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$ live inside $\mathbb{Z}$ and still behave like a group — a subgroup.

Definition $H \subseteq G$ is a **subgroup** ($H \leq G$) if: (i) $e \in H$; (ii) $a, b \in H \Rightarrow ab \in H$; (iii) $a \in H \Rightarrow a^{-1} \in H$.
The **centralizer** of $a \in G$ is $C_G(a) = \{g \in G : ga = ag\}$, and $C_G(a) \leq G$.

Plain English A subgroup is a group living inside a larger group. A centralizer collects all elements that commute with a fixed element.

Problem Show that $3\mathbb{Z} = \{\ldots, -6, -3, 0, 3, 6, \ldots\}$ is a subgroup of $(\mathbb{Z}, +)$.

2. Normalizer and Centre of a Group

Definition **Centre:** $Z(G) = \{g \in G : gx = xg \ \forall\, x \in G\}$.
Normalizer of $H \subseteq G$: $N_G(H) = \{g \in G : gHg^{-1} = H\}$.
Both are subgroups of G.

Plain English Centre = elements commuting with everyone. Normalizer = elements that preserve H under conjugation.

Problem Find $Z(S_3)$. *Result:* $Z(S_3) = \{e\}$.

3. Product Subgroups

Definition For $H, K \leq G$, the **product** is

$$HK = \{hk : h \in H,\ k \in K\}.$$

HK is a subgroup of G if and only if $HK = KH$.

Plain English All possible combinations of one move from H followed by one from K.

Problem In $\mathbb{Z}_6$, let $H = \{0, 2\}$ and $K = \{0, 3\}$. Find $H + K$.

4. Cosets and Lagrange's Theorem

Definition For $H \leq G$ and $g \in G$:
Left coset: $gH = \{gh : h \in H\}$. **Right coset:** $Hg = \{hg : h \in H\}$.

Lagrange's Theorem: If G is finite, then

$$|G| = |H| \cdot [G : H],$$

where $[G : H]$ is the number of distinct left cosets.

Plain English Cosets tile the group into equal-sized blocks. Subgroup size must divide group size.

Problem If $|G| = 24$, what are the possible orders of subgroups of G?

5. Cyclic Groups

Definition G is **cyclic** if $G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ for some $a \in G$.

Plain English A cyclic group is generated by repeatedly applying one element.

Problem Is $\mathbb{Z}_8$ cyclic? Find a generator.

6. Classification of Subgroups of Cyclic Groups

Definition If $G = \langle a \rangle$ with $|G| = n$, then every subgroup of G has the form $\langle a^d \rangle$ for some $d \mid n$, and there is exactly one subgroup of each such order.

Plain English Cyclic groups have completely predictable subgroup structure — one subgroup per divisor of n.

Problem Find all subgroups of $\mathbb{Z}_{12}$.

7. Normal Subgroups

Definition $N \trianglelefteq G$ if $gN = Ng$ for all $g \in G$, equivalently $gNg^{-1} = N$ for all $g \in G$.

Plain English Left and right cosets coincide. Conjugation maps N to itself.

Problem Show that every subgroup of an abelian group is normal.

8. Conjugacy Classes

Definition The **conjugacy class** of $a \in G$ is

$$\mathrm{Cl}(a) = \{gag^{-1} : g \in G\}.$$

The conjugacy classes partition G.

Plain English Elements in the same class play the same structural role under conjugation.

Problem Find the conjugacy classes of S_3.

9. Homomorphisms and Kernels

Definition A map $\phi : G \to H$ is a **homomorphism** if $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G$. Its **kernel** is $\ker \phi = \{g \in G : \phi(g) = e_H\}$, and $\ker \phi \trianglelefteq G$.

Plain English A homomorphism translates group structure; the kernel measures what collapses to the identity.

Problem For $\phi : \mathbb{Z} \to \mathbb{Z}_5$, $\phi(n) = n \bmod 5$, find $\ker \phi$.

10. Quotient Groups

Definition If $N \trianglelefteq G$, the **quotient group** is

$$G/N = \{gN : g \in G\}$$

with operation $(aN)(bN) = (ab)N$.

Plain English Each coset is compressed into a single point; the result is a new, smaller group.

Problem Describe $\mathbb{Z}/4\mathbb{Z}$.

11. Group Presentation

Definition A **group presentation** specifies generators and relations:

$$G = \langle\, a, b \mid a^2 = b^3 = e \,\rangle.$$

Plain English Describe a group by its building blocks and the rules they obey, without listing every element.

Problem Give a presentation of the cyclic group of order 5. *Answer:* $\langle a \mid a^5 = e \rangle$.

12. Free Groups

Definition The **free group** $F(S)$ on a set S consists of all reduced words in elements of S and their formal inverses, with concatenation as the group operation. No relations are imposed beyond the group axioms.

Plain English A free group is a language of letters with no grammar rules. Every group arises as a quotient of some free group.

Problem Is the free group on one generator isomorphic to $\mathbb{Z}$?

Module 2 complete. Three grand tools: **subgroups**, **cosets**, and **homomorphisms**. Together they form the structural toolkit of abstract algebra.

Module 2 — 12 Core Theorems of Subgroup Theory

Master keys of abstract algebra: subgroups, cosets, and homomorphisms.

1. Subgroup Criterion

Statement A nonempty subset $H \subseteq G$ is a subgroup if and only if

$$a, b \in H \implies ab^{-1} \in H.$$

Meaning One clever condition replaces the usual three-part verification.

Problem Show that $H = \{x \in \mathbb{Z} : 5 \mid x\}$ is a subgroup of $(\mathbb{Z}, +)$.

2. Centre of a Group is a Subgroup

Statement The **centre** of G is

$$Z(G) = \{g \in G : gx = xg \ \forall\, x \in G\}.$$

Then $Z(G) \leq G$.

Meaning All elements that commute with everything form a subgroup.

Example In any abelian group, $Z(G) = G$.

3. Centralizer is a Subgroup

Statement For $a \in G$, the **centralizer** is

$$C_G(a) = \{g \in G : ga = ag\}.$$

Then $C_G(a) \leq G$.

Meaning All elements commuting with a specific element a form a subgroup.

Problem Find the centralizer of (12) in S_3.

4. Lagrange's Theorem

Statement If $H \leq G$ and G is finite, then

$$|G| = |H| \cdot [G : H].$$

In particular, $|H|$ divides $|G|$.

Meaning The size of every subgroup must divide the size of the group.

Example $|S_4| = 24$, so possible subgroup orders are $1, 2, 3, 4, 6, 8, 12, 24$.

5. Order of an Element Divides Order of the Group

Statement For any element a of a finite group G,

$$|a| \mid |G|.$$

Meaning An element's cycle length must divide the group size.

Example If $|G| = 15$, possible element orders are $1, 3, 5, 15$.

6. Subgroups of Cyclic Groups

Statement If $G = \langle a \rangle$ with $|G| = n$, then every subgroup of G has the form

$$\langle a^d \rangle, \quad d \mid n,$$

and there is exactly one subgroup of each order dividing n.

Meaning Cyclic groups have completely predictable subgroup structure.

Example $\mathbb{Z}_{12}$ has subgroups of orders $1, 2, 3, 4, 6, 12$.

7. Every Subgroup of a Cyclic Group is Cyclic

Statement If G is cyclic and $H \leq G$, then H is cyclic.

Meaning Cyclic groups are structurally simple — their subgroups inherit the same cyclic form.

Problem Find the generators of the unique subgroup of order 3 in $\mathbb{Z}_{12}$.

8. Normal Subgroup Criterion

Statement $N \trianglelefteq G$ if and only if

$$gNg^{-1} = N \quad \text{for all } g \in G.$$

Meaning Conjugation by any group element maps the subgroup to itself.

Example Every subgroup of an abelian group is normal.

9. Kernel of a Homomorphism is Normal

Statement If $\phi : G \to H$ is a homomorphism, then

$$\ker \phi = \{g \in G : \phi(g) = e_H\} \trianglelefteq G.$$

Meaning Kernels always form normal subgroups.

Problem Find $\ker \phi$ for $\phi : \mathbb{Z} \to \mathbb{Z}_6$, $\phi(n) = n \bmod 6$.

10. First Isomorphism Theorem

Statement If $\phi : G \to H$ is a homomorphism, then

$$G/\ker \phi \cong \phi(G).$$

Meaning A homomorphism compresses a group by collapsing its kernel; the quotient recovers the image exactly.

Example $\mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}_5$.

11. Cosets Partition the Group

Statement The left cosets $\{gH : g \in G\}$ form a **partition** of G. Each coset has exactly $|H|$ elements.

Meaning Cosets divide the group into equal-sized, non-overlapping blocks.

Example If $|G| = 12$ and $|H| = 3$, there are exactly 4 cosets.

12. Quotient Group Existence

Statement If $N \trianglelefteq G$, then the set of cosets

$$G/N = \{gN : g \in G\}$$

forms a group under the operation $(aN)(bN) = (ab)N$.

Meaning Only normal subgroups allow the formation of a well-defined quotient group.

Example $\mathbb{Z}/4\mathbb{Z}$ is a cyclic group of order 4.

The big picture. Module 1 asked *what is a group?* Module 2 asks: what subgroups live inside it, how can we divide it, and how can groups communicate? Three grand tools emerge: **subgroups**, **cosets**, and **homomorphisms**.

Module 3 — Properties of S_n

The kingdom of permutations. Cycles, parity, and the universal language of groups.

1. Properties of S_n

Illustration All rearrangements of $\{1, 2, 3\}$ — there are 6 of them — form the group S_3. Each rearrangement is a permutation.

Definition S_n is the group of all bijections $\{1, \ldots, n\} \to \{1, \ldots, n\}$ under composition.

$$|S_n| = n!$$

Plain English S_n collects every possible rearrangement of n objects. It appears throughout combinatorics, physics, and cryptography.

Problem Find $|S_6|$.

2. Cycle Notation

Illustration The motion $1 \to 3 \to 5 \to 1$ is a loop. We write it compactly as $(1\ 3\ 5)$.

Definition A **k-cycle** $(a_1\ a_2\ \cdots\ a_k)$ maps $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_k \mapsto a_1$ and fixes all other elements. Fixed points are omitted.

Plain English Cycle notation replaces full permutation tables with compact circular motion descriptions.

Problem Write in cycle form the permutation $1 \mapsto 2,\ 2 \mapsto 4,\ 4 \mapsto 1,\ 3 \mapsto 5,\ 5 \mapsto 3$.

3. Even and Odd Permutations

Illustration Every permutation can be built from two-element swaps (transpositions). The number of swaps needed — even or odd — is a fixed property.

Definition A permutation is **even** if it is a product of an even number of transpositions, and **odd** otherwise. Parity is well-defined: no permutation is both.

Plain English Every permutation carries a hidden parity fingerprint that cannot be changed by rewriting.

Problem Is $(1\ 2\ 3)$ even or odd?
Hint: $(1\ 2\ 3) = (1\ 3)(1\ 2)$ — two transpositions, so even.

4. Cycle Decomposition

Illustration A complicated permutation decomposes into independent loops, like a

machine splitting into separate rotating gears.

Definition Every $\sigma \in S_n$ can be written **uniquely** (up to order of cycles) as a product of **disjoint cycles**. The order of σ equals the lcm of its cycle lengths.

Plain English Disjoint cycle decomposition is the natural "factorisation" of any permutation.

Problem Find the cycle decomposition of $1 \mapsto 4,\ 4 \mapsto 3,\ 3 \mapsto 1,\ 2 \mapsto 5,\ 5 \mapsto 2$.

5. Alternating Group A_n

Definition

$$A_n = \{\sigma \in S_n : \sigma \text{ is even}\}, \qquad |A_n| = \tfrac{n!}{2}.$$

$A_n \trianglelefteq S_n$. For $n \geq 5$, A_n is simple (no normal subgroups except $\{e\}$ and A_n itself).

Plain English Exactly half of all permutations are even; they form the alternating group.

Problem Find $|A_5|$.

6. Cayley's Theorem

Statement Every group G is isomorphic to a subgroup of $S_{|G|}$.

Illustration No matter how abstract a group is, it can always be viewed as permutations of its own elements. Permutations are the universal language of groups.

Plain English Every group is a permutation group in disguise.

Problem If $|G| = 5$, Cayley's theorem guarantees G embeds in which symmetric group?

7. First Isomorphism Theorem

Statement If $\phi : G \to H$ is a homomorphism, then

$$G/\ker\phi \;\cong\; \phi(G).$$

Plain English The kernel is what collapses; removing it reveals the true image of the group.

Problem For $\phi : \mathbb{Z} \to \mathbb{Z}_6$, $\phi(n) = n \bmod 6$, identify $\mathbb{Z}/\ker\phi$.

8. Isomorphism Classes of Small Finite Groups

Key Facts

- Order 1: trivial group only.

- Order $2, 3$: cyclic group only.
- Order 4: $\mathbb{Z}_4$ or the Klein four-group $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.
- Order 5: cyclic group only (5 is prime).

Plain English Groups of the same order need not be isomorphic; order 4 already gives two genuinely different structures.

Problem List all groups of order 3 up to isomorphism.

9. Cauchy's Theorem

Statement If p is prime and $p \mid |G|$, then G contains an element of order p.

Plain English Every prime hiding in the size of the group must appear as an element's cycle length.

Problem If $|G| = 14$, show that G contains an element of order 7.
Reason: 7 is prime and $7 \mid 14$.

10. Sylow's Theorem (Statement)

Statement Let $|G| = p^n m$ where p is prime and $p \nmid m$. Then:

(i) G has at least one subgroup of order p^n (a **Sylow p-subgroup**).

(ii) All Sylow p-subgroups are conjugate.

(iii) The number of Sylow p-subgroups divides m and is $\equiv 1 \pmod{p}$.

Plain English Groups always contain large prime-power subgroups, and Sylow's theorem controls exactly how many there are.

Problem For $|G| = 12 = 2^2 \cdot 3$, identify the possible orders of Sylow subgroups.

Module 3 complete. Three dominating ideas: cycle decomposition, parity, and the symmetric and alternating groups. These are foundational for Galois theory and modern physics symmetry.

Module 4 — Rings

Algebraic machines that can both add and multiply. Language shaped by *Emmy Noether*.

1. Examples and Basic Properties of Rings

Illustration The integers $\mathbb{Z}$ allow addition and multiplication; both operations stay inside $\mathbb{Z}$. This is the prototype of a ring.

Definition A **ring** $(R, +, \cdot)$ satisfies:

(i) $(R, +)$ is an abelian group.

(ii) Multiplication is associative: $(ab)c = a(bc)$.

(iii) Distributive laws: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

A ring with $ab = ba$ is **commutative**; one with a multiplicative identity 1 is a **ring with unity**.

Plain English A ring is a number system with two operations. Examples: $\mathbb{Z}$, matrices, polynomials, modular arithmetic.

Problem Verify that $(\mathbb{Z}, +, \cdot)$ is a ring.

2. Subrings and Integral Domains

Definition A subset $S \subseteq R$ is a **subring** if $a, b \in S \Rightarrow a - b \in S$ and $ab \in S$.
A commutative ring with unity is an **integral domain** if

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

(No zero divisors.)

Plain English A subring is a ring inside a ring. An integral domain is an honest number system where a product is zero only if a factor is zero.

Problem Is $\mathbb{Z}_6$ an integral domain?
Answer: No, since $2 \cdot 3 = 0$ in $\mathbb{Z}_6$ but $2 \neq 0$ and $3 \neq 0$.

3. Ideals

Illustration An ideal is a sponge inside a ring: multiplying any ring element with something from the ideal always lands back in the ideal.

Definition A subset $I \subseteq R$ is an **ideal** if:

(i) $a, b \in I \Rightarrow a - b \in I$.

(ii) $r \in R$, $a \in I \Rightarrow ra \in I$ (and $ar \in I$ for two-sided ideals).

Plain English Ideals are the ring-theoretic analogue of normal subgroups: they allow the construction of quotient rings.

Problem Show that $5\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

4. Algebra of Ideals

Definition For ideals I, J of a ring R:

$$I + J = \{a + b : a \in I,\, b \in J\}, \qquad IJ = \left\{ \sum_{\text{finite}} a_i b_i : a_i \in I,\, b_i \in J \right\}.$$

Both $I + J$ and IJ are ideals of R.

Plain English Ideals can be combined by addition and multiplication to build new ideals.

Problem Let $I = 2\mathbb{Z}$ and $J = 3\mathbb{Z}$ in $\mathbb{Z}$. Find $I + J$.
Answer: $I + J = \mathbb{Z}$, since $\gcd(2, 3) = 1$.

5. Quotient Rings

Illustration Collapse every element of an ideal to zero. The remaining cosets form a new, simpler ring.

Definition If I is an ideal of R, the **quotient ring** is

$$R/I = \{a + I : a \in R\}$$

with $(a + I) + (b + I) = (a + b) + I$ and $(a + I)(b + I) = ab + I$.

Plain English Quotient rings are the ring analogue of factor groups — they simplify structure by identifying elements that differ by something in I.

Problem Describe $\mathbb{Z}/3\mathbb{Z}$ and list its elements.

6. Chinese Remainder Theorem

Illustration Knowing $x \equiv 2 \pmod 3$ and $x \equiv 3 \pmod 5$ uniquely pins down x modulo 15.

Definition If m and n are coprime positive integers, then

$$\mathbb{Z}/mn\mathbb{Z} \;\cong\; \mathbb{Z}/m\mathbb{Z} \;\times\; \mathbb{Z}/n\mathbb{Z}.$$

More generally, a system of congruences with pairwise coprime moduli has a unique solution modulo the product of the moduli.

Plain English Split a large modulus problem into independent smaller problems, solve each, then reconstruct the answer.

Problem Solve $x \equiv 1 \pmod 3$, $x \equiv 2 \pmod 5$.
Answer: $x \equiv 7 \pmod{15}$.

Module 4 complete. Three central ideas: subrings, ideals, and quotient rings. Next: prime ideals, maximal ideals, and fields of fractions — the gateway to algebraic geometry and advanced number theory.

Module 5 — Prime & Maximal Ideals

The deep internal hierarchy of rings. Systematized by *Emmy Noether*.

1. Prime Ideals and Maximal Ideals

Prime Ideal An ideal $P \subsetneq R$ is **prime** if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

In $\mathbb{Z}$, $(p) = p\mathbb{Z}$ is prime iff p is a prime number.

Maximal Ideal An ideal $M \subsetneq R$ is **maximal** if the only ideals containing M are M itself and R. It is the largest proper ideal — a ceiling that cannot be raised.

Key Relationship Every maximal ideal is prime. The converse fails in general.

Plain English A prime ideal detects when a product "contains" a factor. A maximal ideal is as large as a proper ideal can get.

Problem Is $2\mathbb{Z}$ a prime ideal in $\mathbb{Z}$? Is $5\mathbb{Z}$ maximal in $\mathbb{Z}$?

2. Quotient Rings by Prime and Maximal Ideals

Key Theorems Let R be a commutative ring with unity and I an ideal.

$$I \text{ is prime} \iff R/I \text{ is an integral domain.}$$

$$I \text{ is maximal} \iff R/I \text{ is a field.}$$

Plain English Dividing by a prime ideal removes zero divisors. Dividing by a maximal ideal produces the most perfect algebraic structure — a field.

Example $\mathbb{Z}/5\mathbb{Z}$ is a field since $5\mathbb{Z}$ is maximal.

Problem Is $\mathbb{Z}/6\mathbb{Z}$ a field? Justify.

3. Ring Homomorphisms

Definition A map $\phi : R \to S$ is a **ring homomorphism** if

$$\phi(a+b) = \phi(a) + \phi(b) \quad \text{and} \quad \phi(ab) = \phi(a)\phi(b)$$

for all $a, b \in R$. The **kernel** is $\ker \phi = \{r \in R : \phi(r) = 0\}$, an ideal of R.

Plain English A ring homomorphism is a structure-preserving translation that respects both addition and multiplication.

Problem For $\phi : \mathbb{Z} \to \mathbb{Z}_5$, $\phi(n) = n \bmod 5$, find $\ker \phi$.

4. First Isomorphism Theorem for Rings

Statement If $\phi : R \to S$ is a ring homomorphism, then

$$R/\ker\phi \;\cong\; \phi(R).$$

Plain English Remove what collapses to zero (the kernel); the quotient is identical in structure to the image.

Problem For $\phi : \mathbb{Z} \to \mathbb{Z}_7$, $\phi(n) = n \bmod 7$, identify $\mathbb{Z}/\ker\phi$.
Answer: $\mathbb{Z}/7\mathbb{Z} \cong \mathbb{Z}_7$.

5. Field of Fractions

Illustration From $\mathbb{Z}$ we build $\mathbb{Q}$ by allowing fractions $\frac{a}{b}$. The same construction works for any integral domain.

Definition For an integral domain R, the **field of fractions** is

$$\mathrm{Frac}(R) = \left\{\frac{a}{b} : a, b \in R,\ b \neq 0\right\}\Big/\sim,$$

where $\frac{a}{b} \sim \frac{c}{d}$ iff $ad = bc$. It is the smallest field containing R.

Plain English The field of fractions adds division to an integral domain, just as $\mathbb{Q}$ adds division to $\mathbb{Z}$.

Problem What is the field of fractions of the polynomial ring $\mathbb{R}[x]$?

The hierarchy. Prime ideal $\to$ Integral domain $\to$ Field of fractions and Maximal ideal $\to$ Field. These relationships are the backbone of modern algebra and algebraic geometry.

Module 6 — Applications of Algebra

The bridge from abstract algebra to science, technology, and cryptography.

1. Cryptography

Illustration Online messages are scrambled using modular arithmetic. The RSA system encodes a message M as $C = M^e \bmod n$ and recovers it via $M = C^d \bmod n$.

Mathematical Idea RSA relies on the difficulty of factoring large numbers and on the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$. Correctness is guaranteed by Euler's theorem:

$$M^{\phi(n)} \equiv 1 \pmod{n}.$$

Plain English Group theory ensures encryption and decryption are inverse operations. Security rests on the hardness of factoring.

Problem Compute $7^4 \bmod 10$.
Solution: $7^2 = 49 \equiv 9$, then $9^2 = 81 \equiv 1$. Answer: 1.

2. Symmetry in Physics

Illustration Rotations of three-dimensional space form the group $SO(3)$. Physical laws that are rotationally invariant are governed by this group.

Mathematical Idea A symmetry group classifies all transformations that leave a physical system unchanged. Noether's theorem connects each continuous symmetry to a conserved quantity (energy, momentum, etc.).

Plain English If a physical object or law looks the same after a transformation, that transformation belongs to a symmetry group.

Problem How many distinct rotational symmetries does a square possess?

3. Error-Correcting Codes

Illustration Data transmitted over the internet or via satellite may be corrupted. Error-correcting codes (e.g. Hamming codes) detect and repair errors automatically.

Mathematical Idea Codes are constructed using vector spaces over finite fields $\mathbb{F}_q$. A linear $[n, k, d]$-code can correct up to $\lfloor (d-1)/2 \rfloor$ errors.

Plain English Algebra allows the design of codes that automatically detect and repair transmission errors.

Problem In $\mathbb{F}_2$, compute $1 + 1$.

4. Computer Graphics

Illustration Every rotation, reflection, and scaling of a 3D object in a video game is performed by multiplying a matrix from $GL(n, \mathbb{R})$.

Mathematical Idea Geometric transformations correspond to elements of matrix groups. Composition of transformations is matrix multiplication.

Plain English Matrix groups are the mathematical engine behind 3D graphics and animation.

Problem Find the determinant of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

5. Chemistry and Molecular Symmetry

Illustration The methane molecule CH_4 has rotational and reflective symmetries forming the tetrahedral group T_d. These symmetries predict molecular vibrations and spectral lines.

Mathematical Idea Each molecule belongs to a **point group** — a finite group of symmetry operations (rotations, reflections, inversions).

Plain English Group theory lets chemists classify and predict how molecules behave under physical transformations.

Problem How many lines of reflective symmetry does an equilateral triangle have?

6. Coding Theory and Data Compression

Illustration QR codes and digital storage use arithmetic over finite fields $\mathbb{F}_p$ to ensure reliable data retrieval.

Mathematical Idea Finite fields $\mathbb{Z}/p\mathbb{Z}$ (for prime p) provide the algebraic foundation for Reed–Solomon codes and similar error-correcting systems used in CDs, DVDs, and deep-space communication.

Plain English Finite field arithmetic enables computers to store and transmit data reliably even in the presence of noise.

Problem Compute $3 + 4 \pmod 5$.

Module	Core idea
1	What is a group?
2	Subgroups, cosets, homomorphisms
3	Groups via permutations (S_n)
4	Rings — addition and multiplication
5	Ideals and the path to fields
6	Applications across science and technology

The course builds the core language of modern algebra, leading toward Galois theory, algebraic number

theory, cryptography, and algebraic geometry.